Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/03/2007
PI Name:	Wiese, Claudia Ph.D.		
Project Title:	A role for homologous recombination in complex DSB repair after HZE particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
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Comments:	For information purposes only--PI moved in June 2014; however, funding did not go with her. As of 6/1/14, Colorado State University, Department of Environmental and Radiological Health Sciences, 485 MRB - 1618 Campus Delivery, Fort Collins, CO 80523-1618, Office: (970) 491 7618, Email: Claudia.Wiese@colostate.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2009, per K. Willison (3/07) NOTE: Question re end date, per JSC; changed back to 9/30/2006 (1/07) NOTE: project extended for full length of proposal, per J. Dardano (2/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Overall Rationale: The overall goal of this proposal is to investigate whether homologous recombinational DNA repair (HRR) contributes to the repair of double-strand breaks (DSBs) generated by the radiation types found in the space radiation environment. This proposal is a continuation of our previous work demonstrating that recombination is induced in human cells exposed to Fe ions. Here, we aim to directly assess the role of HRR in the repair of DNA damage after high linear energy transfer (LET) radiation in mammalian cells, an investigation that has never been carried out before. High LET charged particles deposit large amounts of energy along the ion trajectories, leading to the induction of highly localized DNA damage. These spatially correlated DSBs rejoin with slower kinetics and to less completeness than DSBs induced by low LET radiation. In mammalian cells, X-ray induced DSBs are primarily repaired by non-homologous end joining (NHEJ) in G1, but HRR plays a critical role in S- and G2-phases of the cell cycle. Several reports indirectly suggest that HRR, generally a precise form of DNA repair, plays an important role in the repair of correlated DSBs. Importantly, radiation-induced human tumors arise through a multi-step process of genetic change, and defects in HRR leading to the stimulation of error-prone DNA repair pathways may accelerate this process. For this reason, it is important to investigate directly whether alterations in the ability to perform HRR can sensitize humans to HZE particles. Approach: We will determine in syngeneic human cells whether defects in HRR affect the extent of cell killing and the mechanism of mutagenesis by densely ionizing Fe ions. RNA interference technology will be used to impair HRR (targeting XRCC3, Rad51D or Rad51) in the human lymphoid cell line WTK1. For comparison purposes, NHEJ will also be targeted, and X-rays will be used to test high vs. low LET radiation effects. The Fe particle-induced mutation frequencies will be determined at the autosomal TK1 locus and at the X-linked HPRT locus for parental WTK1 cells and for one representative derivative of WTK1 cells with a 'loss-of-function' phenotype for HRR and for NHEJ. Sets of TK1 mutants will be collected and the Fe ion-induced, X-ray-induced and spontaneous mutation spectra will be compared to discriminate between recombinational and deletional events. Furthermore, we will investigate in hamster and in human mutant cells whether impaired HRR enhances the cytotoxic effects of Fe ions. Wild-type CHO cells and CHO mutant cell lines impaired in HRR (Rad51D, XRCC3) or in NHEJ (DNA-PKcs) will be compared. Both asynchronous and synchronous cell cultures will be used and the relative contributions of both DNA repair pathways to cell survival will be assessed. DSB repair in CHO mutant cell lines and wild-type cells will be measured using a recently described assay that quantifies DSB rejoining by gH2AX foci formation. The effect of the loss of XRCC3 on the cellular sensitivity to Fe particles in an hTERT-immortalized human fibroblast strain will also be determined.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The overall goal of this Research Project is to investigate whether homologous recombinational DNA repair (HRR) contributes to the repair of double-strand breaks (DSBs) generated by the radiation types found in the space radiation environment. We hypothesize that, in some cases, correlated, clustered radiation damage, as induced by heavy charged (HZE) particles, requires the resection of the damaged DNA and the HRR pathway, and therefore is not channeled into the non-homologous end-joining (NHEJ) pathway. We are investigating syngeneic near-normal human cells that are either HRR-proficient or HRR-impaired. In this study, ablation of HRR is induced by gene-specific knockdown using RNA interference. Homologous recombination (HR) is conserved in all organisms and is a prerequisite for both DNA repair and the resumption of stalled replication forks. As replication fork stalling occurs during most cycles of replication, proteins required for HRR generally are essential and cannot easily be deleted in human cells. Therefore, RNA interference is the method of choice for attenuating HRR in near-normal human cells. Importantly, HRR is a DNA repair pathway with close ties to cancer biology and crucial for maintaining genomic stability, limiting mutagenesis and preventing carcinogenesis. It is well established that conditions promoting reduced levels of HRR compromise the fidelity of DSB repair and correlate with an elevated cancer risk. The risk of developing cancer is increased in individuals exposed to space radiation, and defects in HRR, leading to the increased utilization of error-prone DNA repair pathways, are likely to contribute to this process. Therefore, it is a necessity to establish the relevance of HRR to HZE radiation, both for better prediction of the astronaut's sensitivity to radiation carcinogenesis and for proper radiation risk assessment. Using syngeneic human cells, this Research Project is assessing whether defects in HRR affect the extent of cell killing (Aim 1) and the levels and mechanisms of mutagenesis (Aim 2) by densely ionizing Fe ions. RNA interference is used to impair HRR (targeting proteins essential for HRR: XRCC3 and RAD51AP1) in two human lymphoblastoid cell lines of different p53 status (Aims 1&2). The Fe particle-induced mutation frequencies were determined at the autosomal TK locus and at the X-linked HPRT locus for one representative derivative of each cell line with a 'loss-of-function' phenotype for HRR (Aim 2), and these mutation frequencies are compared to control cells transfected with a non-depleting hairpin. In future NSRL runs, sets of TK mutants will be collected and the Fe ion-induced, X-ray-induced and spontaneous mutation spectra will be assessed to discriminate between recombinational and deletional events (Aim 2). As part of Aim 3 synchronized hTERT-immortalized normal human dermal fibroblasts (NHDF) were tested for their ability to localize RAD51 into RAD51 foci in response to Fe ions. HRR-defective Chinese Hamster Ovary (CHO) cells have also been tested for their response to 1 GeV/n Fe ions, investigating both RAD51 foci formation and cell survival. We have conducted two experimental runs at NSRL using 1 GeV/n Fe ions during the last year. We have found that human lymphoblastoid cells depleted for either XRCC3 or RAD51AP1, both proteins essential for HRR, are mildly sensitized (~1.2- to 1.4-fold) to the cytotoxic effects of 1 GeV/n Fe ions. This sensitization is more pronounced for TK6 cells expressing wild type p53 than for WTK1 cells expressing mutant p53. Furthermore, we have determined the levels of spontaneous and radiation-induced mutagenesis in human lymphoblastoid cell lines with silenced XRCC3. As expected, more spontaneous TK and HPRT mutants were detected in TK6 and WTK1 cells expressing low levels of XRCC3, implicating that under conditions that limit DSB repair by HRR, error-prone DSB repair mechanisms (i.e. NHEJ) may elevate spontaneous mutagenesis. Following exposure to 1.2 Gy of 1 GeV/n Fe ions, significantly more HPRT-deficient mutants were recovered from cells with silenced XRCC3 compared to control-depleted cells, suggesting that HRR-mediated repair events or wild-type levels of XRCC3 protein limit Fe ion-induced mutagenesis at HPRT. However, the levels of Fe ion-induced TK mutants did not vary significantly between XRCC3-deficient and control cells, suggesting that the contribution of recombinational events to mutagenesis at TK may be minor or that, due to the impaired levels of HRR in XRCC3-depleted cells, viable TK mutants arising from NHEJ events can not be recovered. Our findings require replication in future NSRL runs.

Unlike for human cells, HRR-knockouts of CHO cells are viable and were tested by us for their response to 1 GeV/n Fe ions. Using colony formation assays, we have found that, compared to CHO wild type cells, RAD51D-deficient CHO cells show enhanced sensitivity (~ 1.5-fold) to the cytotoxic effects of 1 GeV/n Fe ions. Expression of human RAD51D in the RAD51D-knockout reverted cell survival back to wild type levels, supporting the notion that HRR contributes to cell survival after exposure to Fe ions. In wild type, RAD51D-deficent and complemented CHO cell lines we have investigated RAD51 and 53BP1 foci formation in response to 1 Gy of 1 GeV/n Fe ions. In CHO wild type and RAD51D-complemented RAD51D-knockout cells, both 53BP1, a DSB sensor and checkpoint protein, and RAD51, the HRR strand transferase, form microscopically discernible accumulations as early as 20 min after exposure to 1 Gy of 1 GeV/n Fe ions. In HRR-defective RAD51D-knockout cells 53BP1 foci form normally, but RAD51 foci formation is abrogated after exposure to Fe ions, and this defect in localization of RAD51 into focal accumulations is likely to contribute to the enhanced levels of Fe ion-induced cytotoxicity observed for RAD51D-knockout cells. As mentioned above, ectopic expression of human RAD51D in RAD51D knockout cells restores RAD51 foci formation in response to Fe ions.

HR largely relies on the presence of a sister chromatid that serves to provide the template for lesion repair. As a consequence, HRR predominantly occurs in late S- and G2-phases when the sister chromatid is present. We have used hTERT-immortalized human fibroblasts synchronized in G1- or S-phase and investigated whether the formation of RAD51 foci after exposure to Fe ions is cell cycle regulated. We have found that hTERT-immortalized human fibroblasts in G1 are able to form both 53BP1 and g-H2AX foci, but unable to form RAD51 foci in response to Fe ion irradiation. However, when synchronized in S-phase, hTERT-immortalized human fibroblasts show RAD51, 53BP1 and g-H2AX foci after exposure to Fe ions, all of which co-localize along the ion trajectory where the radiation damage induced is expected to be most severe. Our results show that the formation of RAD51 foci in response to Fe ions is cell cycle regulated.

During the course of the last year we have collected evidence showing that RAD51, the strand transferase that is essential for cell viability and the key player in HRR, readily is recruited to DNA lesions induced by 1 GeV/n Fe ions in a cell cycle dependent manner. Compared to X-rays, RAD51 foci following exposure to Fe ions are observed at lower doses and at earlier times. In addition, our cell survival studies and mutation analyses point to the importance of a fully functional HRR pathway in the repair of radiation damage induced by 1 GeV/n Fe ions.

Bibliography Type:	Description: (Last Updated: 04/11/2018)
Abstracts for Journals and Proceedings	Wiese C, Größer T, Kovalenko O, Collins DW, Kratz K, Rydberg B, Schild D. "PIR51/RAD51AP1 is required for the maintenance of genomic stability after DNA damage." 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 5-8, 2006. Abstract Book, 53rd Annual Meeting of the Radiation Research Society, November 2006. , Nov-2006
Abstracts for Journals and Proceedings	Schild D, Kratz K, Wiese C. "A Role for Homologous Recombination in Complex DSB Repair after HZE Particles." NASA Human Research Program Investigators' Workshop, League City, TX, February 12-14, 2007. Abstract Book, Proceedings, NASA Human Research Program Investigators' Workshop, February 2007. , Feb-2007
Abstracts for Journals and Proceedings	Wiese C, Größer T, Collins DW, Rydberg B, Schild D. "The human RAD51AP1/PIR51 protein is required for homologous recombination and genomic stability." 13th International Congress of Radiation Research, San Francisco, CA, July 8-12, 2007. Abstract Book, 13th International Congress of Radiation Research, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Schild D, Seidler S, Zafar F, Wiese C. "A Role for Homologous Recombination in Complex DSB Repair after HZE Particles." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, CA, July 13-15, 2007. Abstract Book, 18th Annual NASA Space Radiation Investigators' Workshop, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Wiese C, Dray E, Größer T, San Filippo J, Shi I, Collins DW, Tsai M-S, Williams G, Rydberg B, Sung P, Schild D. "Promotion of Homologous Recombination and Genomic Stability by RAD51AP1 via RAD51 Recombinase Enhancement." FASEB Summer Research Conference: Genetic Recombination & Genome Rearragements, Snowmass Village, CO, July 28-August 2, 2007. Abstract Book, FASEB Summer Research Conference: Genetic Recombination & Genome Rearragements, Snowmass, CO, July 2007. , Jul-2007
Articles in Peer-reviewed Journals	Wiese C, Dray E, Größer T, San Filippo J, Shi I, Collins DW, Tsai M-S, Williams G, Rydberg B, Sung P, Schild D. "Promotion of Homologous Recombination and Genomic Stability by RAD51AP1 via RAD51 Recombinase Enhancement. " Molecular Cell, In Press, July 2007. , Jul-2007